



Experimental Characterization of a Propulsion System for Multi-rotor UAVs

Daniele Sartori¹ · Wenxian Yu¹

Received: 7 June 2018 / Accepted: 8 February 2019 / Published online: 20 February 2019
© Springer Nature B.V. 2019

Abstract

The propulsion system of a multi-rotor UAV plays a fundamental role in the aircraft flight characteristics. In fact, it generally represents the major contributor to the aerodynamic forces acting on the vehicle. While several approaches for modeling rotor thrust and drag forces exist, the problem of identifying the parameters for these models is still challenging. In this paper we propose a systematic method for identifying a limited number of parameters which guarantee accurate thrust and drag prediction according to Blade Element Theory (BET). Simple experimental tests employing a popular rotor system and a custom-made quadrotor are used both in the identification phase and for the final validation. The discussion of the results illustrates the accuracy of the method, while highlighting the modeling limit of BET. A refinement using Blade Element Momentum Theory is proposed and validated with the support of experimental data.

Keywords UAV · Propulsion · Rotor · Testing · Modeling · Multi-rotor · Quadrotor aircraft · Blade element theory · Blade element momentum theory · Thrust · Drag · Flapping

1 Introduction

The propulsion system is the core element of multi-rotor Unmanned Aerial Vehicles (UAVs). Since this category of aircraft solely relies on the rotors for flight, an accurate modeling of the rotor system is important for appropriate aircraft design and performance evaluation [15].

Rotors are designed to produce thrust. In multi-rotor aircraft, the thrust and the reaction torque of each rotor are adjusted to generate the desired resultant force and moment. These determine the aircraft attitude, flight direction and speed. In detail, the adjustment is performed by modifying the rotor angular speed which acts as control input of the system. Rotors are also responsible for generating a large part of the total vehicle drag, mainly caused by the blades flapping. The rotorcraft static and dynamic behavior strongly depends on the complex function relating the

blade rotational speed to the rotor aerodynamic forces [21]. Accurate modeling of rotor thrust and drag is important for vehicle stability analysis [4], control system design [20] and state estimation [1, 13, 22].

The problem of modeling the rotor forces has been tackled by many works in recent years. However, as explained by [7], the research has mainly focused on thrust prediction, commonly modeled as proportional to the square of the rotor angular speed. While this simple formulation is reliable for hovering or low speed flight, it loses accuracy at higher speed. Furthermore, in this flight regime also the rotor drag becomes relevant. As clearly illustrated in [16], including the drag force in the multi-rotor dynamic analysis provides significant benefits. A solution which guarantees the improvement of the thrust estimation and allows drag prediction is the use of the Blade Element Theory (BET). BET is a standard approach in helicopter literature, see for example [12], which can be easily applied to multi-rotor aircraft. A detailed description of rotor thrust and drag modeling for quadrotors based on Blade Element Theory is proposed in [2]. Within the BET formulation, the modeling of the blade flapping angle plays an important role. This parameter is directly responsible for the flapping drag and it requires a dedicated estimation approach. Formulations for the flapping angle estimation can be found in [2, 8, 12, 19].

✉ Daniele Sartori
dan.sartori@sjtu.edu.cn

¹ Shanghai Key Laboratory of Beidou Navigation and Location Services, Shanghai Jiao Tong University, 800 Dong Chuan Road, Shanghai, 200240, China

A problem in the application of BET to multi-rotor UAVs is the identification of the blade geometric and aerodynamic parameters. Many rotor blades, in particular when dealing with small multi-rotor vehicles, were designed for RC model applications and few technical details are provided. Direct measure is an option, for example in [10] the blade is cut into segments and the characteristic lengths and angles of the airfoil sections are measured. This approach, however, can be technically difficult and introduce inaccuracies, since generally small quantities are involved. Another option consists in the empirical estimation through experimental testing. In [21] the rotor geometric parameters are lumped together with air density into two constants which are used to define the relationship between rotor angular speed and thrust. The value of the constants is determined by test rig and in-flight data analysis. While reliable results are achieved, no details on the blade geometry are obtained and no drag estimation is performed. A comprehensive approach is proposed by [7], the parameters of an enhanced BET model are identified with a complex minimization routine applied to flight data and are verified with wind tunnel experiments.

A particular difficulty exists in defining the coefficients for the flapping angle model. Generally quadrotors have hingeless rotor hub, the blade bends under the action of aerodynamic and inertial forces at a spanwise location assumed to be the equivalent hinge offset. The flapping angle depends on properties such as moment of inertia and stiffness about this virtual hinge. The methods for determining the flapping coefficients can include a simple closed formula for zero flapping hinge offset, [2, 19], or more complex analysis combining the measure of the blade deflection and the study of its natural frequency of vibration, [8, 12].

In this paper we propose an approach to model the rotor system characteristics which allows to accurately evaluate the thrust and drag it produces. As theoretical basis we use the Blade Element Theory as formulated in [12] coupled with the flapping model of [8]. The unknown parameters are evaluated combining direct measures and estimations from experimental data. Experiments include static thrust test and flight test with a quadrotor vehicle, all supported by data collected with a motion tracking system. We choose to model a rotor system widely employed in multi-rotor scientific projects with the idea to support the realization of future research in this field. In fact, in the explanation of the parameters estimation process, we try to illustrate in detail the choices performed and the solutions adopted. Validation results show the ability of this method to create a framework for the estimation of rotor properties which guarantee accurate thrust and drag modeling. We also discuss the limits of the BET model and suggest the upgrade

to the Blade Element Momentum Theory for the inflow ratio calculation. According to the authors' best knowledge this approach represents a novelty in the unmanned multi-rotor field, both for the combination of the BET with the adopted flapping model and for the experimental estimation methodology.

The paper is organized as follows. Section 2 introduces the rotor forces mathematical model, Section 3 describes the experimental approach used to identify the unknown parameters and Section 4 illustrates the validation tests. Finally, conclusions are drawn in Section 5.

2 Rotor Model

We illustrate the rotor aerodynamic model based on Blade Element Theory for a rotating blade in forward flight. The focus is on the main derivation steps, additional details can be found in [2] and [12], from which the formulation is based.

A rotor i moving through the air at velocity V_i and spinning with an angular speed ω_i generates a thrust force T_{ri} and a drag force D_{ri} . The thrust is assumed perpendicular to the rotor plane while the drag lying in the rotor plane and aligned to the velocity direction, see Fig. 1.

A local frame of reference centered in the rotor hub is defined: x_r and y_r lie in the rotor disk with x_r aligned to the airflow velocity, z_r is perpendicular to the rotor disk and points upwards, y_r completes the right handed frame. The azimuth angle Ψ describes the angular position of the blade. Its value is $\Psi = 0^\circ$ when the blade is aligned with the x_r axis and grows to $\Psi = 90^\circ$ when aligned to the y_r axis.

The Blade Element Theory divides the rotor blade into a number of infinitesimal elements. Each of these elements is considered as a quasi-2D airfoil capable of generating forces and moments, see Fig. 2. The infinitesimal forces dF_x , dF_y and dF_z of each element contribute to the generation of the overall rotor thrust and drag, also called H-force. Thrust and drag are obtained by integrating these infinitesimal forces along the blade span dy , which is incorporated

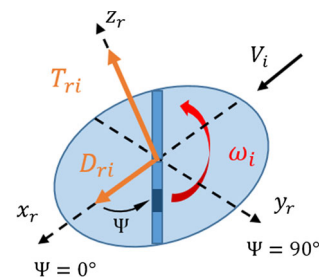


Fig. 1 Forces acting on a moving rotor

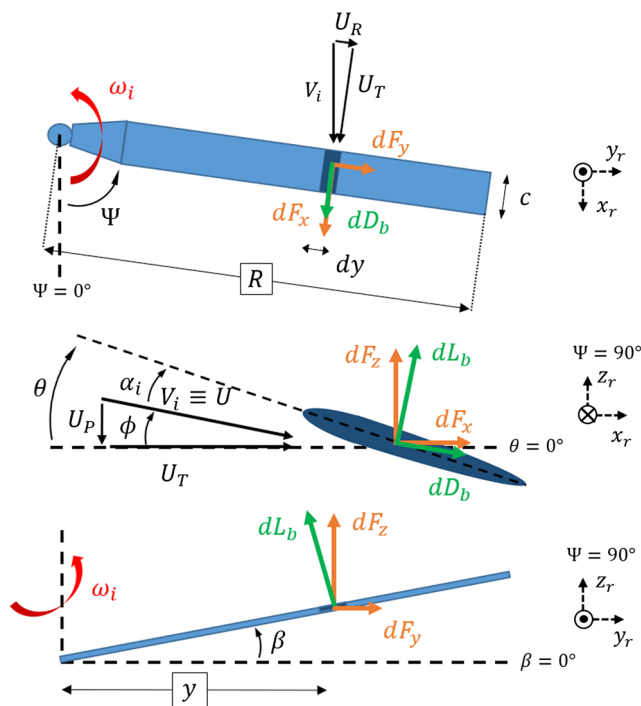


Fig. 2 BET analysis of a spinning rotor blade

within the forces expressions, and over a rotor revolution $d\Psi$,

$$\begin{aligned} T_{ri} &= \frac{N}{2\pi} \int_0^{2\pi} \int_0^R dF_z d\Psi \\ D_{ri} &= \frac{N}{2\pi} \int_0^{2\pi} \int_0^R (dF_x \sin \Psi + dF_y \cos \Psi) d\Psi, \end{aligned} \quad (1)$$

where R is the blade radius and N the number of blades. The infinitesimal forces are defined starting from the infinitesimal lift dL_b and drag dD_b generated by the blade element airfoil,

$$\begin{aligned} dF_x &= dL_b \sin \phi + dD_b \cos \phi \\ dF_y &= -(dL_b \cos \phi - dD_b \sin \phi) \sin \beta \\ dF_z &= (dL_b \cos \phi - dD_b \sin \phi) \cos \beta, \end{aligned} \quad (2)$$

with β flapping angle and $\phi = \arctan(U_P/U_T)$ inflow angle. The airfoil element has chord c , span dy , lift slope coefficients $C_{l\alpha}$ and drag coefficient C_d . The infinitesimal lift dL_b and drag dD_b are modeled as

$$\begin{aligned} dL_b &= \frac{1}{2} \rho U^2 c dy C_{l\alpha} \alpha_i \\ dD_b &= \frac{1}{2} \rho U^2 c dy C_d, \end{aligned} \quad (3)$$

with ρ air density, $U = \sqrt{U_T^2 + U_P^2}$ and $\alpha_i = \theta - \phi$ blade angle of attack; θ is the blade pitch angle. If the drag force and the inflow angle are assumed small, the following simplifications are possible: $\sin \phi \approx \phi$, $\cos \phi \approx 1$, $dD_b \sin \phi \approx 0$, $\phi \approx U_P/U_T$ and $U \approx U_T$. By applying

these simplifications and expressing the definition for α_i , the infinitesimal forces of Eq. 2 can be simplified as

$$\begin{aligned} dF_x &= \frac{1}{2} \rho c C_{l\alpha} \left(\theta U_P U_T - U_P^2 + \frac{C_d}{C_{l\alpha}} U_T^2 \right) dy \\ dF_y &= -\frac{1}{2} \rho c C_{l\alpha} \left(\theta U_T^2 - U_P U_T \right) \sin \beta dy \\ dF_z &= \frac{1}{2} \rho c C_{l\alpha} \left(\theta U_T^2 - U_P U_T \right) \cos \beta dy. \end{aligned} \quad (4)$$

The non-dimensional blade radius is introduced, $r = y/R$, with y spanwise distance of the element from the rotor hub. By replacing the forces expressions of Eq. 4 into Eq. 1, it is possible to write

$$\begin{aligned} T_{ri} &= \frac{N}{2\pi} \frac{\rho R^3 \omega_i^2}{2} \int_0^{2\pi} \int_0^1 C_{l\alpha} c \left[\theta \left(\frac{U_T}{\omega_i R} \right)^2 - \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} \right] \cos \beta dr d\Psi \\ D_{ri} &= \frac{N}{2\pi} \frac{\rho R^3 \omega_i^2}{2} \int_0^{2\pi} \int_0^1 C_{l\alpha} c \left\{ \left[\theta \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} - \left(\frac{U_P}{\omega_i R} \right)^2 \right] \sin \Psi + \right. \\ &\quad \left. - \left[\theta \left(\frac{U_T}{\omega_i R} \right)^2 - \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} \right] \sin \beta \cos \Psi \right. \\ &\quad \left. + \frac{C_d}{C_{l\alpha}} \left(\frac{U_T}{\omega_i R} \right)^2 \sin \Psi \right\} dr d\Psi. \end{aligned} \quad (5)$$

The rotor force coefficients are defined as,

$$\begin{aligned} C_{Tri} &= \frac{T_{ri}}{\rho \pi R^4 \omega_i^2} \\ C_{Dri} &= \frac{D_{ri}}{\rho \pi R^4 \omega_i^2}. \end{aligned} \quad (6)$$

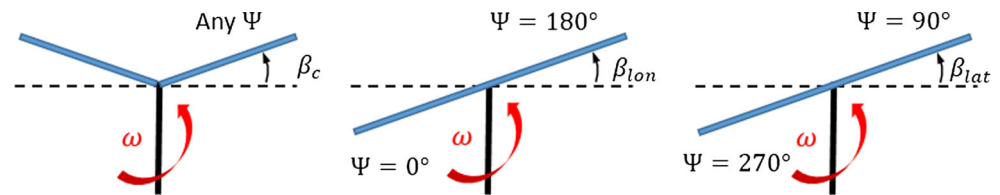
By assuming c and $C_{l\alpha}$ constant along the blade span, Eq. 5 becomes

$$\begin{aligned} C_{Tri} &= \frac{\sigma C_{l\alpha}}{4\pi} \int_0^{2\pi} \int_0^1 \left[\theta \left(\frac{U_T}{\omega_i R} \right)^2 - \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} \right] \cos \beta dr d\Psi \\ C_{Dri} &= \frac{\sigma C_{l\alpha}}{4\pi} \int_0^{2\pi} \int_0^1 \left\{ \left[\theta \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} - \left(\frac{U_P}{\omega_i R} \right)^2 \right] \sin \Psi + \right. \\ &\quad \left. - \left[\theta \left(\frac{U_T}{\omega_i R} \right)^2 - \frac{U_P}{\omega_i R} \frac{U_T}{\omega_i R} \right] \sin \beta \cos \Psi \right. \\ &\quad \left. + \frac{C_d}{C_{l\alpha}} \left(\frac{U_T}{\omega_i R} \right)^2 \sin \Psi \right\} dr d\Psi, \end{aligned} \quad (7)$$

with $\sigma = \frac{Nc}{\pi R}$ rotor solidity, ratio between blade area and disk area. The ratio between the element airflow velocities and the blade tip speed $\omega_i R$ can be expressed as

$$\begin{aligned} \frac{U_T}{\omega_i R} &= r + \mu \sin \Psi \\ \frac{U_P}{\omega_i R} &= \lambda + \mu \sin \beta \cos \Psi + \frac{r \dot{\beta}}{\omega_i}, \end{aligned} \quad (8)$$

Fig. 3 Flapping angle components: coning, longitudinal, lateral



where $\mu = \frac{V_i \cos \alpha_i}{\omega_i R}$ is the rotor horizontal advance ratio and λ is the inflow ratio. The inflow ratio is related to the thrust coefficient by the nonlinear equation

$$\lambda - \mu \tan \alpha_i - \frac{C_{Tri}}{2\sqrt{\mu^2 + \lambda^2}} = 0, \quad (9)$$

an iterative procedure is necessary to solve for λ , T_{ri} and D_{ri} .

A simplified and very handy formulation for the thrust coefficient can be achieved by substituting the expressions of Eq. 8 into Eq. 7 and assuming uniform inflow $\lambda(r, \Psi) = \text{constant}$, constant blade chord and pitch $\bar{\theta}$, small flapping angle β , $\sin \beta \approx \beta$, $\cos \beta \approx 1$. The analytical solution of the C_{Tri} integration is

$$C_{Tri} = \frac{\sigma C_{l\alpha}}{2} \left[\frac{\bar{\theta}}{3} \left(1 + \frac{3}{2} \mu^2 \right) - \frac{\lambda}{2} \right]. \quad (10)$$

The drag formulation includes the blade induced drag, the blade parasitic drag and the rotor flapping drag. The flapping effect, in fact, is a major contributor to multi-rotor vehicles drag. Flapping occurs in forward flight when, due to blade flexibility and unbalance between the advancing and the retreating blades, the rotor plane tilts creating a force opposing the vehicle motion [15]. Accurate descriptions of flapping is found in [1, 2, 8, 19]. The flapping angle β is a function of the blade azimuth,

$$\beta = \beta_c - \beta_{lon} \cos \Psi + \beta_{lat} \sin \Psi, \quad (11)$$

where β_c is the coning angle, β_{lon} the blade longitudinal deflection and β_{lat} the lateral deflection, see Fig. 3.

The system of equations governing flapping motion, according to the model proposed in [8], is

$$\begin{bmatrix} \lambda_\beta^2 & 0 & 0 \\ \frac{\gamma\mu}{6} & 1 - \lambda_\beta^2 & -\frac{\gamma}{8} \\ 0 & \frac{\gamma}{8} & 1 - \lambda_\beta^2 \end{bmatrix} \begin{bmatrix} \beta_c \\ \beta_{lon} \\ \beta_{lat} \end{bmatrix} = \begin{bmatrix} \frac{\gamma\theta}{8} - \frac{\gamma\lambda}{6} \\ 0 \\ \frac{\gamma\mu\theta}{3} \end{bmatrix}, \quad (12)$$

with $\gamma = \rho C_{l\alpha} c R^4 / I_b$ Lock number. The parameter λ_β represents the ratio of the flapping frequency to the angular

rate of the rotor. For stiff propellers, as common in multi-rotors,

$$\lambda_\beta = \sqrt{1 + \frac{3e_f}{2} + \frac{k_\beta}{I_b \omega_i^2}}, \quad (13)$$

where I_b is the effective moment of inertia of the blade about the non-dimensional effective hinge offset $e_f = \frac{4k_\beta}{3N\omega_i^2 I_b}$, k_β is the blade equivalent spring stiffness about e_f .

3 Parameters Identification

The propulsion system analyzed in the current paper is the DJI E310. It is composed of a E420S motor controller, brushless electric motor and DJI 9450 plastic propeller, see Fig. 4. Table 1 illustrates the main properties of the system [6]. This propulsion system is very popular among researchers and has been employed in a variety of multi-rotor projects, see for example [3, 9, 11, 14, 23, 24].

3.1 PWM- ω Identification

The first step of the identification procedure is the study of the relationship between the Pulse Width Modulation (PWM) command and the blade rotational speed ω . The PWM signal is an indicator of the throttle command given by the pilot or by the flight control system. The ESC, according to the input PWM, produces a 3-phase AC output signal which determines the motor spinning speed. The experimental setup for determining the PWM- ω relationship is shown in Fig. 5.

The propulsion system is fixed on a metal structure slightly larger than the motor base in order to minimize aerodynamic disturbances in the propeller downwash. Power is supplied by a 4 cells Li-Po battery pack. The PWM input is manually defined with an RC radio controller and sent through an open-source Pixhawk flight manage-

Fig. 4 DJI E310 propulsion system



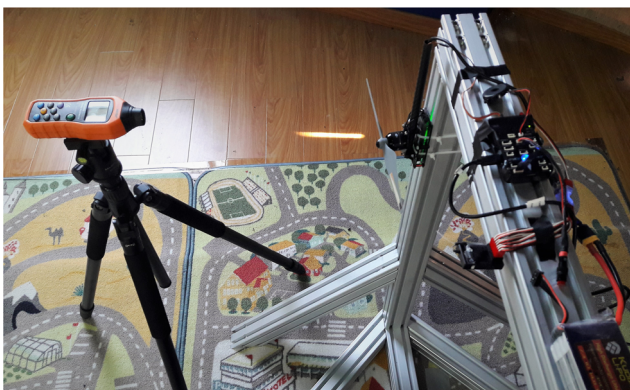
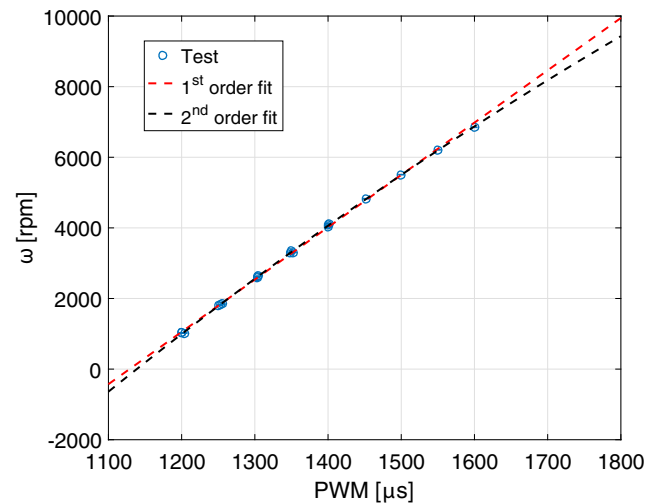
Table 1 DJI E310 parameters

Parameter	Value
ESC max voltage	17.4 V
ESC max current	20 A
Motor KV	960 rpm/V
Motor max thrust @ 12 V	7.85 N
Number of blades	2
Propeller diameter	0.24 m
Propeller thread pitch	0.127 m

ment unit [18]. The Pixhawk is connected to a PC via USB cable, the PWM value is acquired with the MAVROS protocol *rc/out* topic, monitored on screen and saved into a file. The rotor angular speed is measured with an optical tachometer, PeakMeter MS6208B model. This device guarantees contactless measure exploiting laser reflection on a thin metallic strip applied to the rotor blade. The declared accuracy is $\pm 0.03\%$ of the measure ± 2 rpm. Manual PWM steps of about $50 \mu\text{s}$ are given, starting from the minimum rotation value corresponding to about $1200 \mu\text{s}$ and stopping at about $1600 \mu\text{s}$ for safety reasons. About twenty sample points are acquired. The second order relationship $\text{rpm} = -3.21 \cdot 10^3 \text{ PWM}^2 + 23.69 \text{ PWM} - 2.282 \cdot 10^4$ guarantees satisfying fitting, better than a linear relationship for higher PWM values. The second order function guarantees an RMS error of 28 rpm, compared with 53 rpm for the first order one. Results are shown in Fig. 6, the declared error bars fall within the plotted point markers.

3.2 Blade Pitch and Lift Slope Coefficient Identification

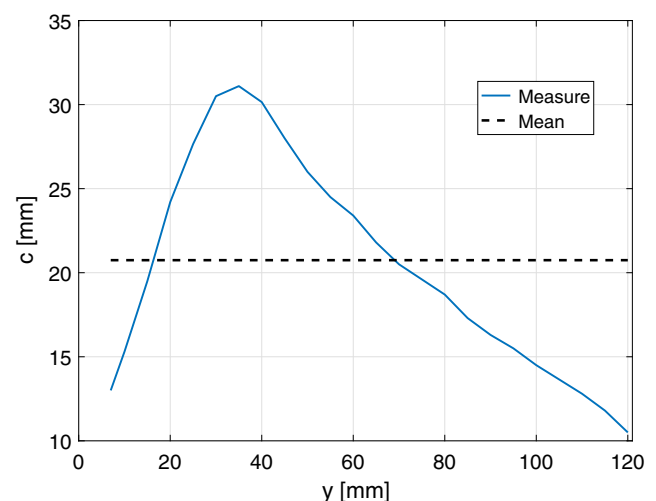
By observing Eqs. 7, 8 and 9 it is possible to identify the blade parameters necessary for the calculation of the rotor thrust and drag: radius R , chord c , pitch θ , lift slope coefficient $C_{l\alpha}$ and drag coefficient C_d . The blade parameters

**Fig. 5** Test setup for determining the PWM- ω relationship**Fig. 6** Experimental PWM - ω relationship

required for the estimation of the flapping angle β , which appears in these equations, will be evaluated in Section 3.3.

The rotor radius $R = 0.12$ m is provided by the manufacturer. The easily measured rotor hub radius $R_{hub} = 0.007$ m is also considered. BET in Section 2 was derived assuming $R_{hub} \ll R$ which is not acceptable in this case. Similarly, in determining (7) the chord c was assumed constant. While this is true for rectangular blades, it does not apply to the DJI 9450 propeller which has some taper. The chord distribution along the blade span is measured with a caliper and it is illustrated in Fig. 7. While formally inaccurate, since in Eq. 7 the chord appears in a product of radius-dependent functions, for sake of simplicity we adopt the mean chord as

$$\bar{c} = \frac{1}{R - R_{hub}} \int_{R_{hub}}^R c dy = 0.0207 \text{ m}, \quad (14)$$

**Fig. 7** Blade chord distribution and mean value

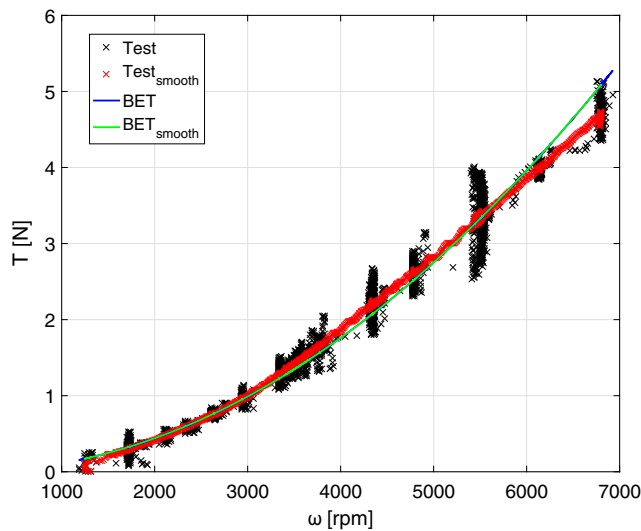


Fig. 10 Measured and BET-fitted thrust force over the rotor angular speed, in the raw and smooth cases

where x_O is the observed input data and y_O is the observed output data. In our case $x = [\bar{\theta}, C_{l\alpha}]$, the rotor speed ω is x_O and the measured thrust T is y_O . The function F is represented by the thrust formulation obtained starting from its coefficient of Eq. 10. Note that using the thrust coefficient from Eq. 7 does not provide significant difference as the test is performed in near-static conditions. An iterative procedure combines Eqs. 10 and 9 solving for C_{Tr} and λ . The measured thrust and the fitting curve through BET are illustrated in Fig. 10 over the rotor angular speed. Operating on the original test data points or on the smoothed ones provides very similar results. The curve guarantees an excellent fit up to 6000 rpm, then it slowly diverges to a maximum error of 0.4 N at 6800 rpm. The effective pitch angle value is estimated as $\bar{\theta} = 0.2547 \text{ rad} = 14.59^\circ$, the lift slope coefficient as $C_{l\alpha} = 5.682 \text{ 1/rad}$.

The blade is treated as a 3D finite wing. The drag coefficient is expressed as $C_d = C_{d0} + k_{in} (C_{l\alpha}\alpha)^2$. The zero-lift drag is related to the Reynolds number $Re = \rho V \bar{c} / \mu_{vis}$ by $C_{d0} = 0.1166 Re^{-0.2}$ [12], where μ_{vis} is the air dynamic viscosity. The value of the induced drag parameter k_{in} is found from the Prandtl's formula, $C_{l\alpha} = C_{l\alpha_{theo}} / (1 + k_{in} C_{l\alpha_{theo}})$, where $C_{l\alpha_{theo}} = 2\pi$ is the theoretical 2D thin-airfoil value for incompressible flows.

3.3 Blade Hinge Offset Identification

The blade parameters necessary to model the flapping motion are the non-dimensional effective hinge offset e_f , the moment of inertia I_b about e_f and the equivalent torsional spring stiffness k_β about e_f .

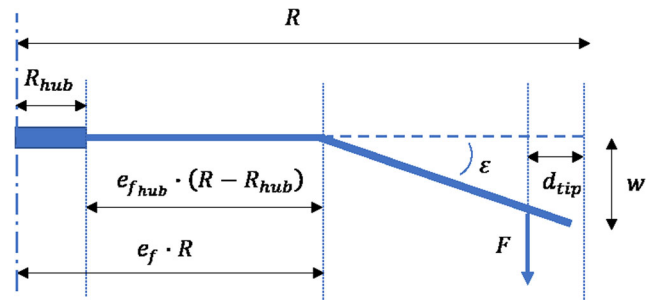


Fig. 11 Blade deflection test schematic approach

The parameters I_b and k_β are expressed as function of the only unknown e_f . The relationship for k_β is determined with a static load test. The blade is horizontally clamped at the hub and a known force $F = 3.4 \text{ N}$ is applied at a distance $d_{tip} = 0.004 \text{ m}$ from the tip, see the scheme in Fig. 11. The force is obtained suspending a weight to a wire passing through a hole drilled in the propeller. The measure of the tip vertical displacement $w = 0.021 \text{ m}$ allows to define the rotation angle ε and, therefore, to express the stiffness k_β and the inertia I_b with the relationships [8]

$$\sin \varepsilon = \frac{w}{(1 - e_{f_{hub}})(R - R_{hub})} \quad (19)$$

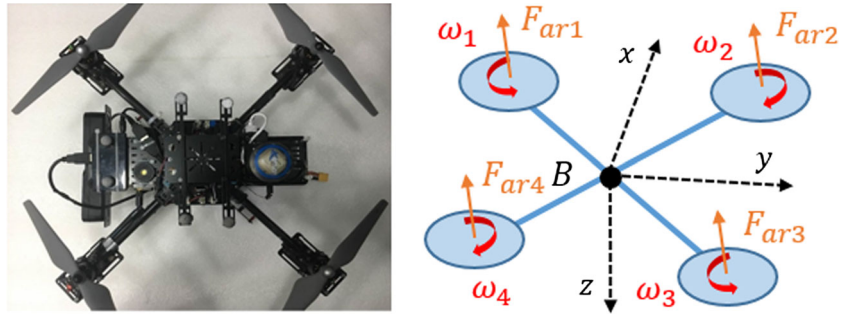
$$k_\beta = \frac{F [(1 - e_{f_{hub}})(R - R_{hub}) - d_{tip}] \cos \varepsilon}{\varepsilon} \quad (20)$$

$$I_b = \frac{4k_\beta}{3N\omega^2 e_{f_{hub}}}, \quad (21)$$

where we refer the effective hinge offset to the part of the blade allowed to flex, $e_{f_{hub}}$. The evaluation of $e_{f_{hub}}$ in helicopter literature [12] requires the estimation of the blade slope under loading, which can be affected by a high degree of uncertainty for the short and stiff blades of quadrotors. The method suggested in [8] uses the measure of the blade vibration natural frequency, which requires complex testing. In this paper we estimate the effective $e_{f_{hub}}$ starting from experimental data gathered during a simple flight. The proposed approach finds the value of $e_{f_{hub}}$ that minimizes the drag error between test data and the BET model prediction.

The test data are collected during an indoor 40 seconds mostly planar flight with our home-made 1.95 kg quadrotor platform, see Fig. 12. It has 6 Vicon markers installed and it uses a Pixhawk flight management unit together with an InvenSense MPU-9250 incorporating 3-axis accelerometer, 3-axis gyroscope and 3-axis magnetometer. The geometric center of the quadrotor B is assumed as its center of mass, see Fig. 12. Here generic body axes $(x, y, z)^T$ have their origin: x and y lie in the airframe plane, with x pointing

Fig. 12 Vehicle used in the tests and quadrotor axis notation



forward and y pointing in the right direction, z complete the right-handed frame by pointing downward. The propeller plane is assumed perpendicular to z . The body axes linear velocity, linear acceleration and angular velocity vectors of the vehicle are called respectively $\mathbf{V} = (v_x, v_y, v_z)^T$, $\mathbf{a} = (a_x, a_y, a_z)^T$ and $\mathbf{\Omega} = (\omega_x, \omega_y, \omega_z)^T$. The total aerodynamic force vector $\mathbf{F}_a = (D_x, D_y, T_z)^T$ is generated by combining the sum of all rotor forces with the quadrotor airframe drag.

The aerodynamic forces from the experimental test are obtained from the accelerometers, multiplying the measured acceleration components with the quadrotor mass m [13],

$$\begin{cases} D_{xt} = ma_x \\ D_{yt} = ma_y \\ T_{zt} = ma_z. \end{cases} \quad (22)$$

The measured test flight conditions are provided as input to the BET model which is evaluated for each of the four rotors. Velocity $\mathbf{V}$ is acquired with Vicon system while the angle of attack is calculated from the body velocity components, $\alpha = \arctan(v_z/v_{xy})$ with v_{xy} velocity in the quadrotor $x - y$ plane. Note that while the proper fixed-wing aeronautical formulation uses v_x at the denominator, here we are not interested in the vertical velocity component with respect to the x axis, but instead we refer to the $x - y$ plane where the vehicle is free to move. Each rotor i faces a velocity $\mathbf{V}_i = \mathbf{V} + \mathbf{\Omega} \times \mathbf{r}_i$, $\mathbf{r}_i$ indicating the location of the rotor with respect to B , and a coherently calculated α_i . The angular velocity $\mathbf{\Omega}$ is provided by the onboard gyroscope. Flow distortion created by the airframe is ignored. The individual PWMs are available from the Pixhawk unit, the rotor angular speeds are evaluated with the PWM- ω relationship previously found. The BET-modeled rotor drag, which is aligned with the total air velocity, is projected on the UAV body axes with an angle $\beta_{ss} = \arcsin(v_y/v_{xy})$ equivalent to fixed-wing aircraft sideslip,

$$\begin{cases} D_{xri} = -D_{ri} \cos \beta_{ss} \text{sign}(v_x) \\ D_{yri} = -D_{ri} \sin \beta_{ss} \\ T_{zri} = -T_{ri}, \end{cases} \quad (23)$$

while the thrust only changes sign. The forces obtained from each rotor are combined in unified drag and thrust,

$$\begin{cases} D_{xr} = \sum_{i=1}^4 D_{xri} \\ D_{yr} = \sum_{i=1}^4 D_{yri} \\ T_{zr} = \sum_{i=1}^4 T_{zri}. \end{cases} \quad (24)$$

The airframe is modeled as a cube with equivalent frontal area S_x and S_y , and drag coefficient $C_{Dbody} = 1.05$ [17]. Its drag contributions, in both x and y directions, are added to the total rotor drag, the modeled quadrotor drag forces are

$$\begin{cases} D_{xm} = D_{xr} + D_{xbody} = D_{xr} + \frac{1}{2} \rho v_x^2 S_x C_{Dbody} \\ D_{ym} = D_{yr} + D_{ybody} = D_{yr} + \frac{1}{2} \rho v_y^2 S_y C_{Dbody}. \end{cases} \quad (25)$$

These operations are performed for a number of e_{fhub} values $\in [0, 1]$. The selected e_{fhub} is the one minimizing the RMS error between the measured and the modeled drag forces over the whole test data period

$$\min_{e_{fhub}} \sqrt{[rms(D_{xt} - D_{xm})]^2 + [rms(D_{yt} - D_{ym})]^2}. \quad (26)$$

The experimental data are also employed to refine the estimation of the blade pitch angle $\bar{\theta}$ previously modeled as constant along the rotor blade radius. By assuming a linear twist and that $\bar{\theta}$ is achieved at 50% of the radius $r \in [R_{hub}/R, 1]$, the new pitch angle expression is

$$\theta = \bar{\theta} + \theta_{tw} (r - 0.5). \quad (27)$$

The twist parameter θ_{tw} is estimated with the nonlinear curve-fitting approach of Eq. 18. In this case $x = \theta_{tw}$, $x_O = \omega_i$, $y_O = T_{zt}$ and the function F is the thrust

formulation obtained from the thrust coefficient of Eq. 7. The overall process is summarized in Algorithm 1.

Algorithm 1 Estimation of $e_{f_{hub}}$.

```

1: Define a vector of  $e_{f_{hub}} \in [0, 1]$ 
2: Define a vector of  $r \in [R_{hub}/R, 1]$ 
3: Define a vector of  $\Psi \in [0, 2\pi]$ 
4: for all  $e_{f_{hub}}$  do
5:   Evaluate  $\delta, k_\beta, I_b$  with Eqs. 19, 20, 21
6:   for all rotors  $i$  do
7:     Evaluate  $\lambda_\beta$  with Eq. 13
8:     while  $err_{convergence} > tolerance$  do
9:       Evaluate  $\beta_c, \beta_{lon}, \beta_{lat}$  with Eq. 12
10:      Find  $\theta_{tw}$  of Eq. 27
11:      for all  $\Psi$  do
12:        Evaluate  $\beta$  with Eq. 11
13:        for all  $r$  do
14:          Evaluate  $\frac{U_T}{\omega_i R}, \frac{U_P}{\omega_i R}$  with Eq. 8
15:        end for
16:      end for
17:      Evaluate  $C_{Tri}, C_{Dri}$  with Eq. 7
18:      Evaluate  $\lambda$  with Eq. 9
19:    end while
20:    Evaluate  $T_{ri}, D_{ri}$  with Eq. 6
21:    Evaluate  $D_{xri}, D_{yri}, T_{zri}$  with Eq. 23
22:  end for
23:  Evaluate  $D_{xm}, D_{ym}$  with Eqs. 24, 25
24: end for
25: Find  $e_{f_{hub}}$  according to Eq. 26

```

The values of $e_{f_{hub}} = 0.66$ and $k_\beta = 0.17$ Nm/rad guarantee the best accordance between measured and BET modeled drag forces, see Fig. 13. It is interesting to

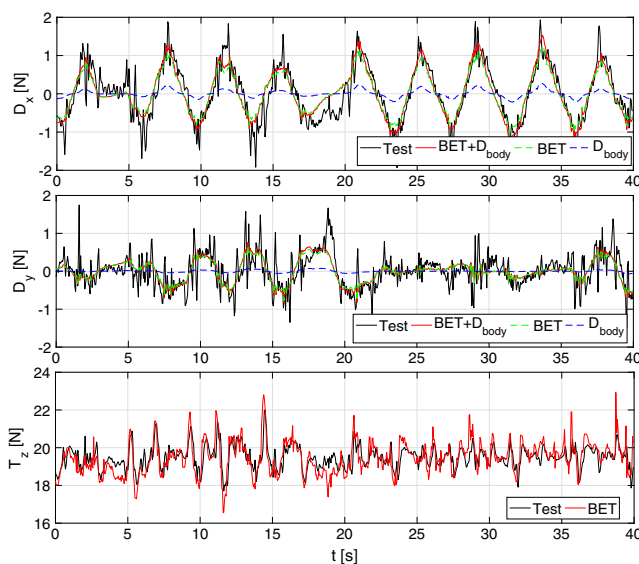


Fig. 13 $e_{f_{hub}}$ identification: test forces and modeled forces

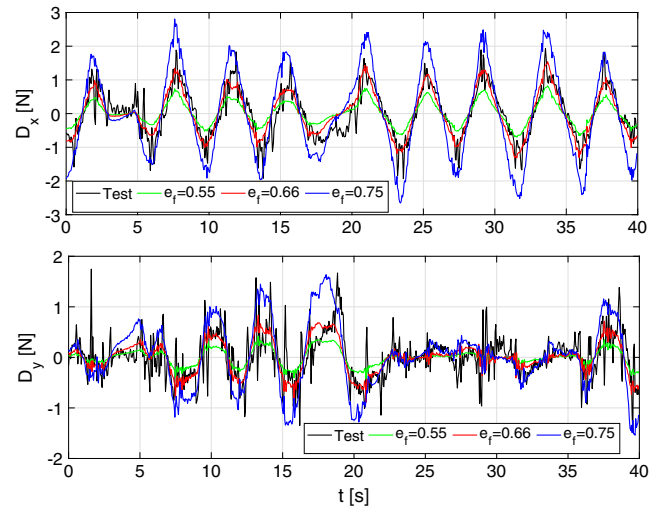


Fig. 14 Test drag and $e_{f_{hub}}$ effect on modeled drag force

observe how the rotor drag is the major contributor to the total quadrotor drag. The airframe, even if characterized with a very penalizing cube aerodynamic model, accounts on average for 9.4% of the full drag on the faster flying x direction and 5.1% on the slower y direction. A good match is also obtained for the thrust force. Despite the value of $e_{f_{hub}}$ appears to be higher than expected (hinge offset are normally very close to the rotor hub), this result is compatible with what observed during the static load test and with the features of the blade. In fact, the DJI 9450 propeller is very stiff and thick near the rotor hub. It is interesting to observe in Fig. 14 the effect that different locations of $e_{f_{hub}}$ have on the drag estimation. For values lower than $e_{f_{hub}} = 0.66$ the drag is underestimated, while for higher values the force value rises quickly.

The blade twist is identified as $\theta_{tw} = 0.0616$ rad = 3.53° . Note that employing the estimation routine of Eq. 18, with a fitting between the BET-modeled thrust and the accelerometer-derived thrust of Eq. 22, could also be used to estimate $\bar{\theta}$ and $C_{l\alpha}$. The methods of Section 3.2 and Section 3.3 are interchangeable for the identification of the blade effective pitch and lift slope coefficient. In this case, these parameters were estimated with a static test and the flight test was used to refine the blade pitch characterization.

4 Model Validation and Refinement

The identified parameters are inserted in the BET mathematical model presented in Section 2. The model is processed with experimental data input (V, α, ω) from a real flight and compared with the forces estimated from

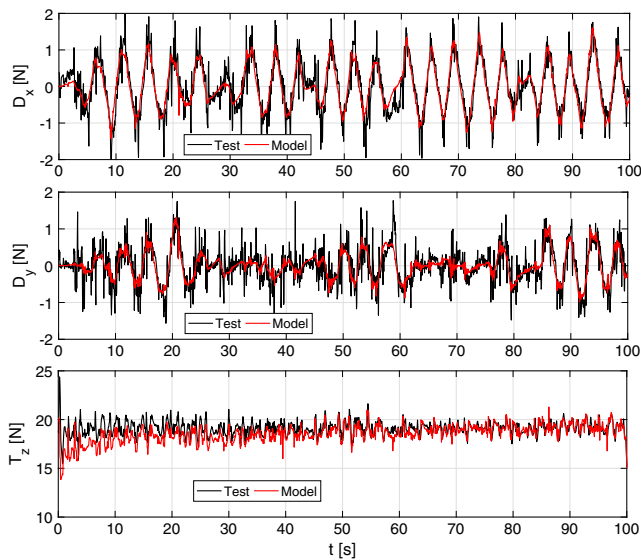


Fig. 15 Validation: test forces and modeled forces over time

accelerometers. The results presented in Figs. 15, 16, 17 illustrate the aerodynamic forces over time and their dependency over rotor angular speed and flight velocity.

An overall satisfying matching is achieved for all forces, in particular for the drag. The RMS error is found to be 0.342 N for D_x and 0.352 N for D_y . In Fig. 17 it is interesting to observe how the linear relationship between lateral velocity and drag, typical of multirotor vehicles, is correctly captured by the model. The BET model, while correctly estimating the thrust at mid-high rpm values, diverges from the measured data at low rotor angular speed, where it underestimates the force. The reason for this is found in the uniform inflow assumption which originates the

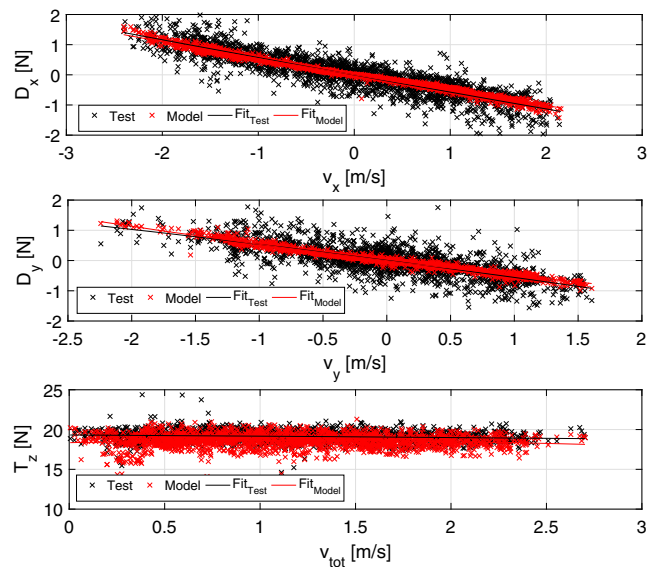


Fig. 17 Validation: test forces and modeled forces over speed

relationship of Eq. 9. In fact, the uniform inflow formulation can not properly capture the aerodynamic forces distribution along the blade span, in particular their span-wise variation due to the changing airflow velocity caused by the rotor angular rotation. The BET thrust RMS error is 0.53 N.

In order to refine the model we adopt the inflow estimation proposed by the Blade Element Momentum Theory (BEMT). This theory combines Momentum Theory and Blade Element Theory, stating the equivalence between

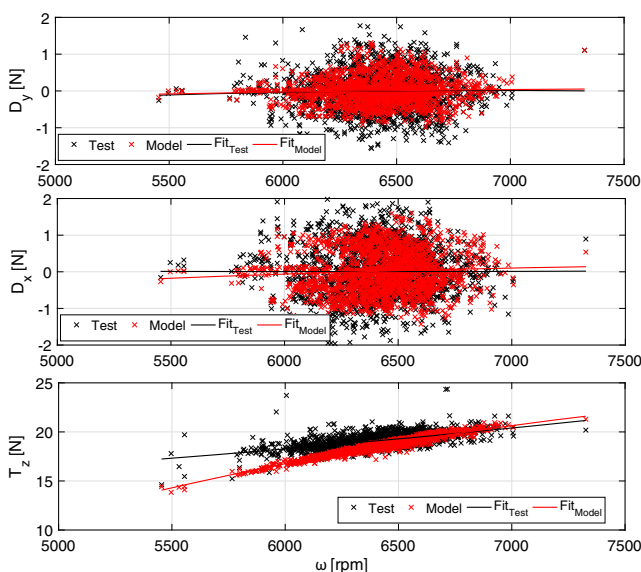


Fig. 16 Validation: test forces and modeled forces over rpm

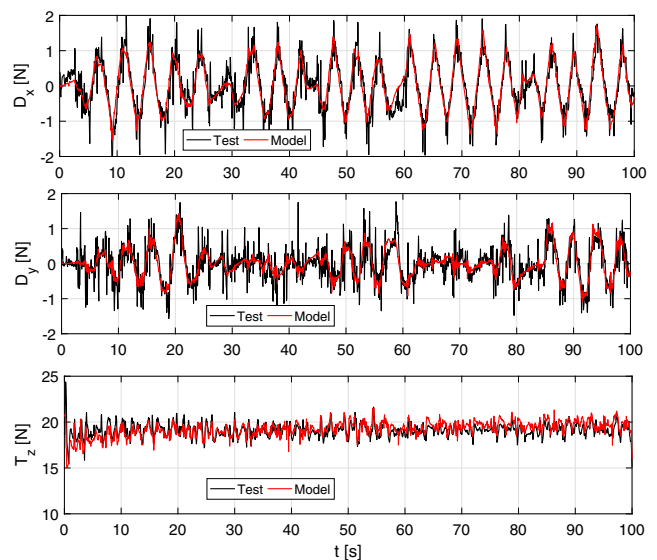


Fig. 18 Validation: test forces and modeled forces over time with the use of BEMT for the inflow calculation

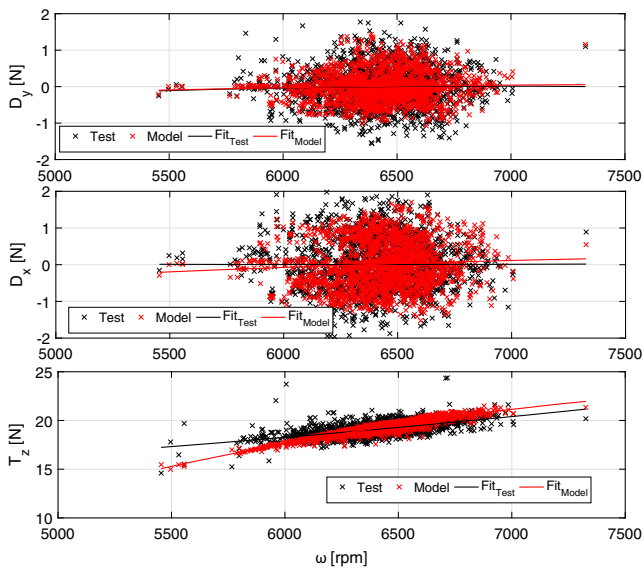


Fig. 19 Validation: test forces and modeled forces over rpm with the use of BEMT for the inflow calculation

circulation and momentum theory of lift [12]. Starting from this assumption, the inflow radial distribution is evaluated as

$$\lambda_{BEMT} = \sqrt{\left(\frac{\bar{\sigma} C_{l\alpha}}{16} - \frac{\lambda_c}{2}\right)^2 + \frac{\bar{\sigma} C_{l\alpha}}{8} \theta r} - \left(\frac{\bar{\sigma} C_{l\alpha}}{16} - \frac{\lambda_c}{2}\right) \quad (28)$$

with $\lambda_c = \frac{-V_i \sin \alpha_i}{\omega_i R}$ rotor vertical advance ratio. Note that similar considerations on the inflow motivated the choice of combining BET and BEMT in the work of [7]. The approach of Section 3.3 is repeated using Eq. 28 instead of Eq. 9. The results of Fig. 18, and in particular of Fig. 19, show an improvement in the thrust estimation over the low rpm range, now the RMS error drops to 0.5 N, with a minimal overestimation at higher rpm values. The drag estimation remains basically unchanged, RMS error 0.34 N for D_x and 0.35 N for D_y , still providing excellent results. Similar values of $e_{f_{hubBEMT}} = 0.67$ and $k_{\beta BEMT} = 0.16$ Nm/rad are found, indicating how the bending properties of the blade can be generally considered independent from the forces model. As expected, the blade pitch twist changes, $\theta_{twBEMT} =$

Table 2 DJI E310 properties summary

Parameter	Value
PWM-rpm	$\text{rpm} = -3.21 \cdot 10^3 \text{ PWM}^2 + 23.69 \text{ PWM} - 2.28 \cdot 10^4$
$\bar{c}$	0.0207 m
$\bar{\theta}$	0.2547 rad
$C_{l\alpha}$	5.682 1/rad
$e_{f_{hub}}$	BET: 0.66; BEMT: 0.67
k_{β}	BET: 0.17 Nm/rad; BEMT: 0.16 Nm/rad
θ_{tw}	BET: 0.061 rad; BEMT: 0.102 rad

0.102 rad = 5.84°. A summary of the major parameters identified with the approach presented in the paper is visible in Table 2.

5 Conclusions

In the present paper a procedure for determining the main blade parameters for thrust and drag modeling has been presented. While the theoretical foundation is the Blade Element Theory, the parameters have been identified through basic aerodynamic modeling and experimental testing. The experiments carried out are straightforward and do not require complex equipment, their realization has been illustrated in detail. The selected rotor system is a popular choice among researchers dealing with unmanned multi-rotor vehicles. As a result, the proposed approach and the obtained results can be used as starting point for a variety of projects focusing on the design and modeling of rotorcraft systems.

The results show the validity of the approach in guaranteeing a satisfying model of the thrust and drag forces. The most critical parameter in the identification process is the equivalent blade hinge offset e_f . Its estimated value is higher than what commonly found in traditional hingeless helicopter blades. Nevertheless, considering the adopted blade shape, its root stiffness and the observed results from the static tests, it is not surprising to find a value above 50% of the radius. Furthermore, e_f can account for the uncertainties existing in other modeled or measured parameters. While it might be not physically accurate, as the equivalent denomination indicates, it guarantees an equivalence between the modeling formulation and the experimental data, with excellent results in the drag estimation.

The thrust estimation based on BET has shown to have difficulties in representing the trend of the experimental data for low rotor angular speeds. In particular, when the rotors are spinning under 6200 rpm the model considerably underestimates the thrust force. The proposed solution is the use of the Blade Element Momentum Theory for the estimation of the rotor inflow, since it provides a more accurate representation of this variable along the blade span.

Future works include the extension and the validation of the proposed approach to other rotor systems and vehicles. In order to extend its validity to high speed flight, it is necessary to accurately model the drag of the aircraft body to clearly separate its contribution from the one of the rotors. In the proposed paper, the low speed regime guarantees, as demonstrated, that also with a penalizing airframe drag characterization the contribution of the rotors is dominant. Finally, it can be useful to extract from the built model an accurate set of maps correlating the main input variables

with the rotor thrust and drag for fast computation within simulators, control systems or state estimators.

Acknowledgments The authors wish to thank Ruocheng Yao, Weiqi Liu and Rongzhi Wang for the support in the realization of the experimental tests.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

References

1. Abeywardena, D., Kodagoda, S., Dissanayake, G., Munasinghe, R.: Improved state estimation in quadrotor mavs: a novel drift-free velocity estimator. *IEEE Robot. Autom. Mag.* **20**(4), 32–39 (2013)
2. Bangura, M., Melega, M., Naldi, R., Mahony, R.: Aerodynamics of rotor blades for quadrotors. [arXiv:1601.00733](https://arxiv.org/abs/1601.00733) (2016)
3. Bazin, J.M., Fields, T.D., Smith, A.J.: Feasibility of in-flight quadrotor individual motor thrust measurements. In: *AIAA Atmospheric Flight Mechanics Conference* (2016)
4. Bristeau, P.J., Martin, P., Salaün, E., Petit, N.: The role of propeller aerodynamics in the model of a quadrotor UAV. In: *European Control Conference*, pp. 683–688 (2009)
5. Capello, E., Park, H., Tavora, B., Guglieri, G., Romano, M.: Modeling and experimental parameter identification of a multicopter via a compound pendulum test rig. In: *2015 Workshop on Research, Education and Development of Unmanned Aerial Systems* (2015)
6. DJI: DJI E310 propulsion system. <https://www.dji.com/e310> (2018)
7. Gill, R., D'Andrea, R.: Propeller thrust and drag in forward flight. In: *IEEE Conference on Control Technology and Applications* (2017)
8. Hoffmann, G.M., Huang, H., Waslander, S.L., Tomlin, C.J.: Quadrotor helicopter flight dynamics and control: theory and experiment. In: *AIAA Guidance Navigation and Control Conference and Exhibit* (2007)
9. Hung, L.N., Bon, L.S.: A quadcopter-based auto cameraman system. In: *2016 (IEEE) Virtual Conference on Applications of Commercial Sensors (VCACS)*, pp. 1–8 (2016)
10. Khan, W., Nahon, M.: Toward an accurate physics-based uav thruster model. *IEEE/ASME Trans. Mechatron.* **18**(4), 1269–1279 (2013)
11. Lai, W.Y., Er, M.J., Ng, Z.C., Goh, Q.W.: Semi-autonomous control of an unmanned aerial vehicle. In: *2016 14Th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, pp. 1–4 (2016)
12. Leishman, J.G. *Principle of Helicopter Aerodynamics*, 2nd edn. Cambridge University Press, Cambridge (2006)
13. Leishman, R.C., Macdonald, J.C., Beard, R.W., McLain, T.W.: Quadrotors and accelerometers: State estimation with an improved dynamic model. *IEEE Control. Syst.* **34**(1), 28–41 (2014)
14. Loianno, G., Spurny, V., Thomas, J., Baca, T., Thakur, D., Hert, D., Penicka, R., Krajnik, T., Zhou, A., Cho, A., Saska, M., Kumar, V.: Localization, grasping, and transportation of magnetic objects by a team of (MAVs) in challenging desert-like environments. *IEEE Robotics and Automation Letters* **3**(3), 1576–1583 (2018)
15. Mahony, R.E., Kumar, V., Corke, P.: Multirotor aerial vehicles: modeling, estimation, and control of quadrotor. *IEEE Robot. Autom. Mag.* **19**(3), 20–32 (2012)
16. Martin, P., Salaün, E.: The true role of accelerometer feedback in quadrotor control. In: *IEEE International Conference on Robotics and Automation*, pp. 1623–1629 (2010)
17. McCormick, B.W. *Aerodynamics, Aeronautics, and Flight Mechanics*, 2nd edn. Wiley, New York (1994)
18. Meier, L., Tanskanen, P., Heng, L., Lee, G.H., Fraundorfer, F., Pollefeys, M.: PIXHAWK: a micro aerial vehicle design for autonomous flight using onboard computer vision. *Auton. Robot.* **33**, 21–39 (2012)
19. Omari, S., Hua, M.D., Ducard, G., Hamel, T.: Nonlinear control of VTOL UAVs incorporating flapping dynamics. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 2419–2425 (2013)
20. Pounds, P., Mahony, R.E., Corke, P.: Modelling and control of a large quadrotor robot. *Control. Eng. Pract.* **18**(7), 691–699 (2010)
21. Powers, C., Mellinger, D., Kushleyev, A., Kothmann, B., Kumar, V.: Influence of aerodynamics and proximity effects in quadrotor flight. *Experimental Robotics Springer Tracts in Advanced Robotics* **88**, 289–302 (2013)
22. Sartori, D., Zou, D., Pei, L., Yu, W.: A revisited approach to lateral acceleration modeling for quadrotor UAVs state estimation. In: *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5711–5718 (2018)
23. da Silva, L.C.B., Bernardo, R.M., de Oliveira, H.A., Rosa, P.F.F.: Multi-UAV agent-based coordination for persistent surveillance with dynamic priorities. In: *2017 International Conference on Military Technologies (ICMT)*, pp. 765–771 (2017)
24. Vacek, L., Atter, E., Rizo, P., Nam, B., Kortvelesy, R., Kaufman, D., Das, J., Kumar, V.: sUAS for deployment and recovery of an environmental sensor probe. In: *2017 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 1022–1029 (2017)

Daniele Sartori received the B.Sc., M.Sc., and Ph.D. degrees from the Polytechnic University of Turin, Italy, in 2006, 2008 and 2014 respectively. After three years of working in the industry as Test Data Analysis Engineer, he is currently with the School of Electronic, Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, where he is Assistant Professor at the Shanghai Key Laboratory of Navigation and Location based Services.

Wenxian Yu received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1985, 1988, and 1993, respectively. From 1996 to 2008, he was a Professor with the College of Electronic Science and Engineering, National University of Defense Technology, where he served as the deputy dean of the school and assistant director of the National Key Laboratory of Automatic Target Recognition. In 2008, He joined the School of Electronics, Information, and Electrical Engineering, Shanghai Jiao Tong University, Shanghai. He served as the executive dean of the School from 2009 to 2011. Currently, he is a Distinguished Professor under the Yangtze River Scholar Scheme, vice dean of the Advanced Industrial Technology Research Institute, and dean of the Academy of Information Technology and Electrical Engineering.